

Metro Ticket Generation System

Phase 1 — Requirement Analysis & Planning | ServiceNow

Document Purpose	Define the approved blueprint before ServiceNow development begins.
Phase Objective	Identify business goals, functional scope, stakeholders, assumptions, risks, and implementation roadmap.
Primary Outcome	A clear, testable and stakeholder-approved foundation for later design and development.

1. Phase Overview

Phase 1 establishes what the Metro Ticket Generation System must achieve before tables, forms, flows, scripts, notifications or security controls are implemented. The focus is requirement discovery, clarification, prioritization and planning.

Workstream	Key Question	Output
Business Objectives	Why are we building it?	Business goals & success criteria
Functional Scope	What should it do?	Features & business rules
Stakeholder Mapping	Who is involved?	Roles & responsibilities
Execution Roadmap	How will we build it?	Phased implementation plan

2. Brainstorming — Initial System Vision

Passenger → Journey Details → Validation → Fare Calculation → Review → Confirmation → Ticket Generation → Notification → Ticket History

Key brainstorming questions:

- Who can book and is login mandatory?
- What passenger/journey information is mandatory?
- How are stations, routes and fares maintained?
- Is fare based on distance, station count or a configurable fare matrix?
- Can tickets be cancelled/refunded/reused?
- What happens for same source/destination or inactive stations?
- Who can view, modify, cancel or administer tickets?
- Which capabilities belong in MVP versus future scope?

3. Business Objectives

The system should automate metro ticket generation while improving consistency, user experience, traceability and administrative control.

Objective	Expected Business Value
Automate ticket generation	Reduce manual processing and generate a unique ticket after successful booking.
Reduce errors	Use validation and centralized business rules to prevent incorrect bookings and fare calculations.
Improve user experience	Provide a short, predictable flow from journey entry to confirmation.
Centralize metro data	Maintain stations, routes, fare rules, bookings and tickets in one ServiceNow application.
Maintain ticket history	Enable authorized users to search and audit booking/ticket records.
Enable administration	Allow authorized administrators to maintain master data and configuration.

4. Functional Scope

The MVP should cover the complete booking-to-ticket lifecycle.

Function	Requirement
User access	Identify authorized passengers, operators and administrators.
Journey booking	Capture passenger, source, destination, journey date, passenger count and ticket type.
Validation	Validate mandatory fields, active stations, different source/destination, date rules and passenger limits.
Fare calculation	Determine applicable fare and calculate total fare using configured rules.
Ticket generation	Create a unique ticket number and store booking/ticket details.
Ticket lifecycle	Support agreed statuses such as Draft, Confirmed, Used, Cancelled or Expired.
Notifications	Send booking confirmation and, if required, cancellation/status notifications.
Search & tracking	Locate tickets using ticket number and selected journey attributes.
Administration	Manage stations, routes, fare rules, statuses and access.

5. Conceptual Booking Flow

Enter Details → Validate → Calculate Fare → Review Summary → Confirm → Create Booking → Generate Ticket → Notify Passenger

6. Business Rules & Brainstorming Outcomes

Area	Rule / Decision to Finalize
Source	Required; must reference an active configured station.
Destination	Required; must reference an active station and differ from source.
Journey date	Past dates should be rejected; same-day/advance-booking policy must be confirmed.
Passenger count	Must be at least 1; maximum allowed count must be defined.
Fare	Use a configurable model rather than hardcoding fare values where practical.
Duplicate booking	Repeated submissions should not create unintended duplicate tickets.
Ticket number	Every successful booking must have a unique identifier.
Cancellation	Eligibility, cutoff, refund and charges must be explicitly defined if included.
Access	Passengers access their own records; administrative configuration requires elevated roles.

7. Edge Cases & Expected Behavior

Scenario	Expected Result
Source empty	Reject booking and show validation message.
Destination empty	Reject booking and show validation message.
Same source/destination	Reject journey as invalid.
Inactive/non-existent station	Reject selection/booking.
Past journey date	Reject according to date policy.
Passenger count = 0	Reject booking.
Passenger count above limit	Reject and show permitted range.
No applicable fare	Prevent ticket generation and surface configuration issue.
Already cancelled ticket	Prevent a second cancellation.
Expired ticket	Prevent use according to validity rules.

8. Suggested ServiceNow Data Model

The conceptual model is agreed in Phase 1 and refined during application design.

Entity	Purpose	Key Examples
Metro Station	Station master	Name, code, active status
Metro Route	Route/network relationship	From, to, distance/order
Metro Fare	Configurable fare rules	Rule, amount, effective status
Metro Booking	Passenger journey request	Passenger, journey, fare, status
Metro Ticket	Generated travel ticket	Ticket number, booking, validity, status
Passenger	Passenger profile/reference	Name, contact/reference ID

9. Stakeholder Mapping

Stakeholder	Responsibilities / Interest
Passenger	Books tickets, reviews journey/fare details, views ticket and uses eligible cancellation functions.
Metro Operator	Views bookings/tickets, monitors operational status and handles booking-related issues.
Metro Administrator	Maintains stations, routes, fares, configuration and administrative access.
Business Owner	Defines policy, approves requirements, confirms scope and accepts delivered functionality.
ServiceNow Developer	Builds application components, automation, validation, integrations and technical logic.
ServiceNow Administrator	Supports platform configuration, roles, environments and deployment controls.
QA / Test Engineer	Creates test cases, validates rules, performs regression and security-oriented testing.

10. Responsibility Matrix

Activity	Business	Admin	Developer	QA
Requirements	A/R	C	C	C
Fare policy	A/R	C	C	C
Station configuration	A	R	C	I
Application development	C	C	R	I
Testing	C	C	C	R
UAT approval	A/R	C	C	R
Production deployment	A	R	R	C

R = Responsible • A = Accountable • C = Consulted • I = Informed

11. Non-Functional Requirements

- Performance: Booking and ticket generation should complete within an agreed response time.
- Security: Access must be role-based; sensitive administrative functions require authorization.
- Reliability: Prevent unintended duplicate ticket creation and preserve transaction consistency.
- Auditability: Important record changes should be traceable through ServiceNow history/audit.
- Maintainability: Fare and business configuration should be configurable where feasible.

12. MVP, Out of Scope & Roadmap

MVP — Must Have

- Station selection and source/destination validation
- Journey date and passenger-count validation
- Fare calculation
- Booking creation and unique ticket generation
- Ticket status and ticket viewing
- Confirmation notification

Should Have

- Ticket cancellation and eligibility rules
- Ticket search/reporting
- Administrator station and fare management

Future / Out of Initial Scope

- Real-time train tracking and external transit APIs
- Online payment gateway and financial settlement
- QR-code gate scanning / smart-card integration
- Dedicated mobile application, SMS or advanced push integrations
- AI-based route optimization or dynamic pricing

13. Execution Roadmap

Phase	Focus	Major Deliverables
Phase 1	Requirements & planning	Approved requirements, scope, stakeholders, rules and roadmap
Phase 2	Application & data model	Scoped app, tables, fields, relationships, forms, roles
Phase 3	Booking implementation	Booking UI/catalog experience, validation and record creation
Phase 4	Business logic & automation	Fare engine, ticket generation, lifecycle and duplicate prevention
Phase 5	Security & notifications	ACLs/roles, notifications and access validation
Phase 6	Testing & UAT	Functional, negative, boundary, security tests and UAT sign-off
Phase 7	Deployment	Production release, training, support handover and documentation

14. Key Risks & Mitigation

Risk	Impact	Mitigation
Unclear fare policy	High	Obtain business approval before coding fare logic.
Changing requirements	Medium	Baseline scope and obtain requirement sign-off.
Duplicate tickets	High	Design unique identifiers and duplicate-prevention logic.
Incorrect access	High	Define roles/ACL expectations early and test them.
Poor master data	Medium	Validate station/fare configuration and active status.
Integration dependency	High	Confirm external API/payment requirements early or exclude from MVP.

15. Assumptions & Phase 1 Exit Criteria

Key assumptions

- ServiceNow is the primary application platform.
- Station and fare master data can be maintained in ServiceNow.
- Fare rules are expected to be configurable.
- Every successful booking produces one unique ticket number.
- Past journey dates are not accepted unless business policy states otherwise.
- Administrative operations are restricted to authorized roles.
- Payment, live tracking and advanced mobile capabilities are not part of the initial MVP unless separately approved.

Phase 1 Exit Checklist

- Business problem and objectives approved
- Functional and non-functional requirements documented
- Stakeholders and responsibilities identified
- MVP and out-of-scope items agreed
- Fare model and ticket lifecycle agreed
- Business rules and edge cases reviewed
- Initial data model agreed
- Security and notification expectations identified
- Risks and assumptions documented
- Execution roadmap approved
- Business/UAT owner provides requirement sign-off

16. Example End-to-End Scenario

Input	Example
Passenger	Rohith
Source	Ameerpet
Destination	Hitech City
Journey Date	10-Sep-2026
Passengers	2

Processing: Validate journey → determine fare → calculate total → create booking → generate unique ticket number → set status → send confirmation.

Example Output: METRO0001001 | Ameerpet → Hitech City | 10-Sep-2026 | 2 passengers | Total fare as configured | Confirmed

17. Phase 1 Final Outcome

Phase 1 is complete when the project has a clear, testable and approved blueprint for the Metro Ticket Generation System. The next phase can then move into ServiceNow application creation, data-model implementation, forms, roles and configuration without relying on unverified assumptions.

Approval Area	Status
Business Objectives	Pending stakeholder sign-off
Functional Scope	Pending stakeholder sign-off
Business Rules / Fare Policy	Pending confirmation
MVP Scope	Pending confirmation
Execution Roadmap	Ready for approval